Higher National Diploma in Engineering (Civil)
04th Year, First Semester Examination - 2019
CE3223– Water Supply and Wastewater Treatment

Instructions for Candidates:
Answer five questions only

No. of questions : 06
No. of pages : 03
Time: 03hours

01. Water has been called the "universal solvent", and it is the fundamental requirement of all life.71% of the Earth's surface is water-covered But less than 1% of it is drinkable (non-saline water) we can avoid the water scarcity by better water management system.
    (i) Briefly discuss about the global/local water issue? (04 marks)
    (ii) Explain the way of managing the water in the Earth? (04 marks)
    (iii)How does global warming affect the water scarcity in the future? (04 marks)
    (iv)Briefly describe the important stages of hydraulic cycle. (04 marks)
    (v) Briefly explain about the living machine technology. (04 marks)
    (Total 20 marks)

02. The major loss is that due to frictional resistance of the pipe, which depends on the inside roughness of the pipe. The common formula for calculating the loss of head due to friction is Darcy's one. (moody diagram is attached in the last page)
    (i) What are the assumptions used in Darcy equation? (04 marks)
    (ii) Derive the Darcy's formula for friction loss of head in a pipe? (05 marks)
    (iii)A pipe 01 m diameter and 10 km long transmits water of velocity of 01 m/sec. The friction coefficient of pipe is 0.006. Calculate the head loss due to friction? (04marks)

(iv) A galvanized iron pipe with a roughness height of $5x10^{-6}$ m with a diameter of 0.05 m and a length of 100m carries a discharge of 0.003 m^3/s. calculate the head loss? (for water : density 1000 kg/m^3 and dynamic viscosity 0.001 kg/ms)

(07 marks)

(Total 20 marks)

03. Water is a very important resource for our environment and there is a strong likelihood that the world will face a fresh water shortage in the near future.

(i) Why do we treat wastewater? (03 marks)

(ii) What are the substances inherent in wastewater creating adverse impacts and what do they result in if not controlled? (05 marks)

(iii) Biological treatment by efficient use of microorganisms is very suitable for Sri Lanka. Do you agree with this sentence and explain your answer? (06 Marks)

(iv) Briefly explain the stabilization pond with a diagram? (06 Marks)

(Total 20 marks)

04.

(i) Briefly describe the Antifoams, Algaecides & Biocides and chemicals which are used in water treatment? (03 marks)

(ii) Explain a suitable process to remove the oil and grease from the wastewater? (03 marks)

(iii) What is the main difference between BOD and COD in the water treatment? (04 marks)

(iv) What are the uses of measurement of BOD in wastewater engineering and briefly explain why five days is used in BOD measurement. (05 marks)

(v) How does pH and temperature affect the coagulation in water treatment? (05 marks)

(Total 20 marks)

05. Draw a Schematic diagram of a typical wastewater treatment plant and briefly describe the purpose of each of the unit processes? (Total 20 marks)

06. Briefly describe the following terms

(i) Fecal coli forms

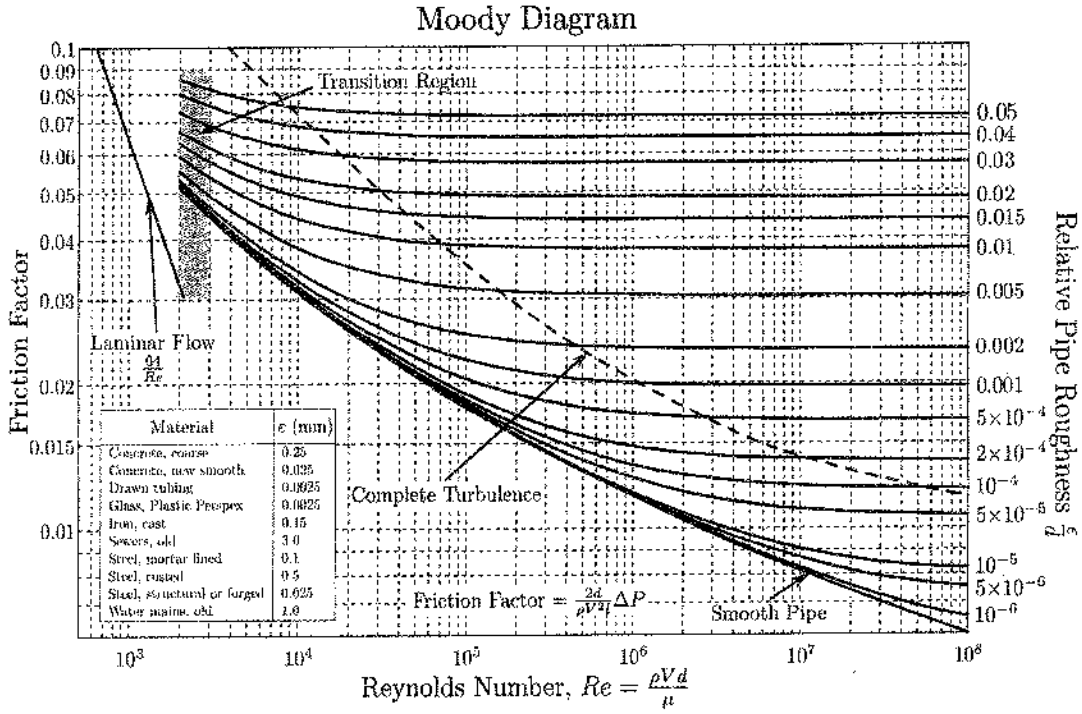
(ii) Constructed wetland

(iii)Eutrophication

(iv)Anaerobic Bacteria

(v) Activated sludge (5 x 04 marks)

(Total 20 marks)



Material	ε (mm)
Concrete, coarse	0.25
Concrete, new smooth	0.025
Drawn tubing	0.0025
Glass, Plastic Perspex	0.0025
Iron, cast	0.15
Sewers, old	3.0
Steel, mortar lined	0.1
Steel, rusted	0.5
Steel, structural or forged	0.025
Water mains, old	1.0